PennState College of Engineering

HAROLD AND INGE MARCUS DEPARTMENT OF INDUSTRIAL AND MANUFACTURING ENGINEERING

Research

NeurMHSP

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Overview

Overview

- Complete to read chapter 1 in ’Introduction to Stochastic programming’. (done)
- Clarify the EMPIRE model mathematical formulation (done)
- Understand the method of scenario generation. (not yet)

Today work

European Commission's Climate Change Response

- **Goal**: Reduce EU greenhouse gas emissions by 80–95% from 1990 levels by 2050.
  - Almost complete decarbonization of the power sector required.
  - Emphasis on large-scale deployment of wind and solar power.
- **Challenges**: Intermittency (something that starts, then stops, then starts up again) and balancing supply-demand due to wind and solar.
- **Importance**: Consider short-term uncertainties in planning investments in generation, transmission, and storage.

Transforming the EMPIRE model

EMPIRE model

- **EMPIRE** (**E**uropean **M**odel for **P**ower system **I**nvestment with **R**enewable **E**nergy) is a *stochastic programming* model, developed to handle the challenges related to intermittent energy production and stochastic energy demand in a long-term investment model.

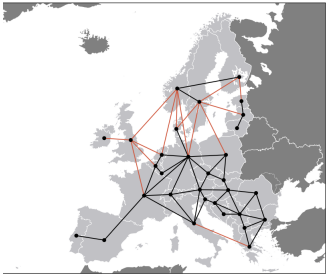

Figure: Spatial detail of the EMPIRE model

- A total of 31 European countries, connected through 55 interconnectors.
- Capacity expansion model designed to assess optimal capacity investments and system operation over **40–50 years.**
- Central planner perspective (i.e. government control, not companies)

Objective

- Minimize the sum of investment and (expected) operational costs for whole of the system, over all time periods.

Approach

- Multi-Horizon Stochastic Programming; Kaut et al. (2014)
  - Why? : To avoid the curse of dimensionality when modeling short-term uncertainty in a long-term model
  - How? : Simultaneously handles short-term dynamics, short-term uncertainty as well as long-term dynamics.

Dynamics

1. long-term dynamics: multiple investment periods
2. short-term dynamics: multiple sequential operational decision periods and market clearing
3. short-term uncertainty: multiple scenarios for input data describing operating conditions
   (wind, solar and load profiles, hydro power production limits, etc.)

Uncertainty

- **Strategic uncertainty**: Assumed perfect foresight in terms of the strategic data. (No uncertainty in strategic decision)
- **Operational uncertainty**:
  - load profiles
  - wind and solar generation profiles
  - seasonal availability of water stored in reservoir for hydroelectric production.

simple picture needed to explain EMPIRE model setting.

Mathematical Formulation

Objective Function

- Minimize the sum of investment(x) and expected operational(y_ω) costs for the system

$$\min_{\mathbf{x},\{y_\omega\}_{\omega\in\Omega}} z = \sum_{i\in\mathcal{I}}(1+r)^{-5(i-1)}\times$$
$$\left\{\sum_{g\in\mathcal{G}} c_{gi}^{\text{gen}} x_{gi}^{\text{gen}} + \sum_{l\in\mathcal{L}} c_{li}^{\text{tran}} x_{li}^{\text{tran}} + \sum_{b\in\mathcal{B}}\left(c_{bi}^{\text{storPW}} x_{bi}^{\text{storPW}} + c_{bi}^{\text{storEN}} x_{bi}^{\text{storEN}}\right)\right.$$
$$\left.+\mathcal{V}\sum_{\omega\in\Omega}\pi_\omega\sum_{s\in\mathcal{S}}\alpha_s\sum_{h\in\mathcal{H}}\sum_{n\in\mathcal{N}}\left[\sum_{g\in\mathcal{G}_n} q_{gi}^{\text{gen}} y_{ghi\omega}^{\text{gen}} + q_{ni}^{\text{ll}} y_{nhi\omega}^{\text{ll}}\right]\right\}$$

, where $\mathcal{I}$ denotes time periods, r denotes discounted rate, $\mathcal{G}, \mathcal{L}$ and $\mathcal{B}$ denote set of generators, interconnectors and storages respectively.

Constraints for operational variables

1. Energy Balance Constraints
2. Capacity balance constraints
3. Maximum production constraints
4. Production for thermal generators constraints
5. Storage energy balance constraints
6. Storage energy limit constraints
7. Hydroelectric power generation
8. Exchange capacity limitation

1. Net Energy Balance

- The sum of net local generation and net import is required to balance the **net load** at every node (n), hour (h), period (i).

$$\sum_{g\in\mathcal{G}} y^{\text{gen}}_{ghi\omega} + \sum_{b\in\mathcal{B}_n} \eta^{\text{discharg}}_{b} y^{\text{discharg}}_{bhiw} - y^{\text{charg}}_{bhi\omega} + \sum_{a\in\mathcal{A}^{in}_n} \eta^{\text{tran}}_{a} y^{\text{flow}}_{ahi\omega} - \sum_{a\in\mathcal{A}^{out}_n} y^{\text{flow}}_{ahi\omega} = \xi^{\text{load}}_{\mathbf{nhi}\omega} - \mathbf{y}^{\mathbf{ll}}_{\mathbf{nhi}\omega}$$

, where $n \in \mathcal{N}$, $h \in \mathcal{H}$, $i \in \mathcal{I}$ and $\omega \in \Omega$

- η: efficiency parameter
  - $\eta^{\text{discharg}}_{b}, \eta^{\text{tran}}_{a} \in (0, 1)$

2. Installments Capacity balance

- Capacity balance for generator, interconnector, storage (power and energy).
- For instance, the capacity of generator at the beginning of period i;

$$v_{gi}^{gen} = \bar{x}_{gi}^{gen} + \sum_{j=i'}^{i} x_{gj}^{gen}, \quad i' = \max\{1, i - \lfloor i_g^{\text{life}}/5 \rfloor\}, i \in \mathcal{I}, g \in \mathcal{G}$$

- v_{gi}^{gen}: maximum installed capacities for generator ($g \in \mathcal{G}$) at the beginning of period i
- $\bar{x}_{gi}^{gen}$: existing capacity at the beginning of period i

We can balance for interconnector,storage (power and energy) in the same way.

How can we interpret those balance equation?

- $i_g^{\text{gen}} = 18$ years
- $\mathcal{I} = \{1, 2, 3, \ldots, 8\}$, $i \in \mathcal{I}$ denote five year time block.

Then, $i^{'} = max\{1, i - \lfloor 18/5 \rfloor\} = max\{1, i - 3\}$, So until $i = 4$, $i^{'}$ is to be 1.

$$v_{g1}^{gen} = \bar{x}_{g1}^{gen} + \sum_{j=1}^{1} x_{gj}^{gen}$$

$$v_{g2}^{gen} = \bar{x}_{g2}^{gen} + \sum_{j=1}^{2} x_{gj}^{gen}$$

$$\vdots$$

$$v_{g4}^{gen} = \bar{x}_{g4}^{gen} + \sum_{j=1}^{4} x_{gj}^{gen}$$

$$v_{g5}^{gen} = \bar{x}_{g5}^{gen} + \sum_{j=2}^{5} x_{gj}^{gen}$$

More specifically,

$$
\begin{aligned}
(i=1)\ \ & v_{g1}^{gen} = \bar{x}_{g1}^{gen} + (x_{g1}^{gen}) \\
(i=2)\ \ & v_{g2}^{gen} = \bar{x}_{g2}^{gen} + (x_{g1}^{gen} + x_{g2}^{gen}) \\
& \vdots \\
(i=4)\ \ & v_{g4}^{gen} = \bar{x}_{g4}^{gen} + (x_{g1}^{gen} + x_{g2}^{gen} + x_{g3}^{gen} + x_{g4}^{gen}) \\
(i=5)\ \ & v_{g5}^{gen} = \bar{x}_{g5}^{gen} + (x_{g2}^{gen} + x_{g3}^{gen} + x_{g4}^{gen} + x_{g5}^{gen})
\end{aligned}
$$

When $i = 5$ (e.g. 21-25 years), x_{g1}^{gen} is disappeared. Because x_{g1}^{gen} is over $i_g^{\mathrm{gen}} = 18$ when it comes to $i = 5$.

3. Maximum production constraints

- Intermittent production, such as wind power and solar power, the availability parameters are stochastic.
- Production from each generator ($\mathcal{G}$) for a hour ($\mathcal{H}$) is limited by the available installed capacity.

$$y^{\text{gen}}_{ghi\omega} \leq \xi^{\text{gen}}_{ghi\omega} v^{\text{gen}}_{gi}$$

, where $g \in \mathcal{G}, h \in \mathcal{H}, i \in \mathcal{I}, \omega \in \Omega$

- $\xi^{\text{gen}}_{ghi\omega}$: Availability parameters
  - $\xi^{\text{gen}}_{ghi\omega} \in (0, 1)$

4.Ramp up of production for thermal generators

- limited to a certain share of the installed capacity

$$|y^{\text{gen}}_{ghi\omega} - y^{\text{gen}}_{g(h-1)i\omega}| \leq \gamma^{\text{gen}}_{g} v^{\text{gen}}_{gi}$$

, where $g \in \mathcal{G}^{\text{Thermal}}, s \in \mathcal{S}, h \in \mathcal{H}^{-}_{s}, i \in \mathcal{I}, \omega \in \Omega$

$$-\gamma^{\text{gen}}_{g} v^{\text{gen}}_{gi} \leq y^{\text{gen}}_{ghi\omega} - y^{\text{gen}}_{g(h-1)i\omega} \leq \gamma^{\text{gen}}_{g} v^{\text{gen}}_{gi}$$

5. Storage energy balance constraints

- At every hour (h), the storage end-level (w) is set as the difference between the energy level in the previous hour ($h-1$) minus the net discharge.

$$w^{\text{stor}}_{b(h-1)i\omega} + \eta^{\text{chrg}}_{b} y^{\text{chrg}}_{bhi\omega} - y^{\text{dischrg}}_{bhi\omega} = w^{\text{stor}}_{bhi\omega}$$

, where $b \in \mathcal{B}, s \in \mathcal{S}, h \in \mathcal{H}^{-}_{s}, i \in \mathcal{I}, \omega \in \Omega$

- $w^{\text{stor}}_{bhi\omega}$: decision variables keep track of the energy level for storage $b \in \mathcal{B}$

6. Energy limit

- The stored energy and charging/discharging are limited by the energy and power installed capacities.

$$
\begin{aligned}
w^{\text{stor}}_{b(h-1)i\omega} &\leq v^{\text{stor}}_{bi} \\
y^{\text{chrg}}_{bhi\omega} &\leq v^{\text{storPW}}_{bi} \\
y^{\text{dischrg}}_{bhi\omega} &\leq \rho_b v^{\text{storPW}}_{bi}
\end{aligned}
$$

, where $b \in \mathcal{B}, h \in \mathcal{H}, i \in \mathcal{I}, \omega \in \Omega$

- ρ_b: non-negative parameter, determines a fixed discharge to charge power capacity ratio.

In simple terms, ρ_b represents the relationship between how fast a battery can be charged and how fast it can discharge its energy.

For example:

- $\rho_b = 1$
  - It means that the maximum charging power and the maximum discharging power of the battery are the **same**.
  - In other words, the time it takes to fully charge the battery would be the same as the time it takes to fully discharge it.
- $\rho_b = 0.5$
  - It means that the maximum charging power is **only half** of the maximum discharging power.
  - In this case, it would take twice as long to fully charge the battery compared to fully discharging it.

7. Hydroelectric power generation

- Water availability in reservoirs and hydroelectric resource potential.
- For regulated hydroelectric generators, energy limits constrain the total production over each season.

$$\sum_{h \in \mathcal{H}_s} y^{\text{gen}}_{ghi\omega} \leq \xi^{\text{RegHydroLim}}_{gsi\omega}$$

- For every node, both regulated and unregulated is limited in terms of annual energy production.

$$\sum_{s \in \mathcal{S}} \alpha_s \times \sum_{h \in \mathcal{H}_s} \sum_{g \in \mathcal{G}^{\text{Hydro}}_n} y^{\text{gen}}_{ghi\omega} \leq \xi^{\text{HydroLim}}_{ni\omega}$$

8. Exchange capacity limitation

- EMPIRE has a simplified network description, only considering import/export links between countries.
- Exchange is limited by the (symmetric) capacity for each interconnector.

$$y^{\text{flow}}_{ahi\omega} \leq v^{\text{tran}}_{li}$$

Constraints for strategic variables

1. limits on possible investments per period for a given technology in a given node

$$\sum_{g \in \mathcal{G}_{tn}} x_{gi}^{\text{gen}} \leq \bar{X}_{ti}^{\text{gen}}$$

, where $t \in \mathcal{T}^{\text{AggTech}}, n \in \mathcal{N}, i \in \mathcal{I}$

2. maximum installed capacity constraints

$$\sum_{g \in \mathcal{G}_{tn}} v_{gi}^{\text{gen}} \leq \bar{V}_{ti}^{\text{gen}}$$

, where $t \in \mathcal{T}^{\text{AggTech}}, n \in \mathcal{N}, i \in \mathcal{I}$

- lines and storages (power and energy) capacity investments and installed capacities

$$v_{bi}^{\mathrm{storPW}} - \beta_b v_{bi}^{\mathrm{storEN}} = 0$$

, where $b \in \mathcal{B}^+, i \in \mathcal{I}$

- $\mathcal{B}^+$: subset of storages, such as batteries
- β_b: fixed storage power to energy ratio.

Future Work

Future works

- Study the stochastic programming with a book 'Introduction to Stochastic Programming' (continuously)
- Understand the method of scenario generation. (continuously)

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